

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	41.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Burning urination;Pelvic pain

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9000.0	cells/μL	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.1	mg/dL	3.5 - 7.2
Blood Urea	25.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clindamycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 400 mg IV infused over 30 minutes daily for a 70 kg adult)

Precautions: Renal function is normal (creatinine 0.7 mg/dL) - no dose adjustment needed. Monitor serum CPK weekly for myopathy, especially if concomitant statin therapy is used. Watch for hypersensitivity reactions. Not recommended in pregnancy or lactation. Ensure adequate hydration; adjust dose if creatinine clearance falls <30 mL/min.

Rationale: Enterococcus faecium is predicted to be resistant to vancomycin (VRE) and only sensitive to daptomycin in the provided list. Daptomycin provides rapid bactericidal activity against VRE and is the appropriate choice given the lack of other susceptible agents.

Antibiotic Drug History

Clinical Summary

Infection Summary:

- **Type:** Recurrent lower-urinary-tract infection presenting as urethritis.
- **Predicted pathogen:** *Enterococcus faecium* (Gram-positive cocci), with a high likelihood of vancomycin resistance (VRE).

Antibiotic susceptibility (predicted):

- **Resistant:** Clindamycin (and presumptively vancomycin).
- **Sensitive:** Daptomycin - the only agent listed as active.

Recommended therapy:

- **Drug:** Daptomycin 6 mg/kg IV once daily (e.g., 400 mg infused over 30 min for a 70-kg adult).

- **Rationale:** Daptomycin is bactericidal against VRE and is the sole predicted susceptible option, making it the appropriate choice for this *E. faecium* infection.

Key precautions & monitoring:

- Renal function is normal (creatinine 0.7 mg/dL); no dose adjustment required now, but reduce dose if creatinine clearance falls <30 mL/min.
- Obtain baseline and weekly serum CPK; monitor for myopathy, especially if the patient is on a statin.
- Watch for hypersensitivity reactions.
- Contraindicated in pregnancy and lactation; ensure effective contraception if applicable.
- Maintain adequate hydration during therapy.

Actionable plan: Initiate daptomycin as above, monitor renal function, CPK, and clinical response, and reassess culture results when available to confirm susceptibility.